

Emergent Einstein equations without an emergent Newton constant: a no-go for sealed-record gravity

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preprint, not peer reviewed, version 2026-06-14. Every load-bearing line is tagged by epistemic status: **[DERIVED]** = established (symbolic or high-precision numeric), with its conditional gates named; **[NO-GO]** = a proved obstruction; **[OBSTRUCTED]** = blocked in practice, with the obstruction characterized; **[OPEN]** = an unsolved obligation. All numeric claims were verified at mpmath `dps` 80 (commonly 100) and symbolically (sympy) where exact; the receipt script is `code/v6_pXI_sealed_record_gravity_nogo_receipts.py`.

Abstract

Within the sealed-records program — physics as whole histories of committed records rather than instantaneous states — we give the gravity sector its sharp, two-sided verdict. **The records derive gravity’s equation of state, and prove they cannot fix its scale.** On the positive side, taking the horizon temperature $T = a/2$ as established elsewhere in this batch (Euclidean-cone regularity and Bisognano–Wichmann modular structure), we show the horizon record entropy is structurally **pure-area** — a curvature-free Wald–Noether charge of the Einstein sector ($d_A/dR = 0$ by construction) — so that the Clausius relation $Q = TS$ and the modular first law $S = K$ close, yielding the Einstein field equations *in form*. The derivation reaches the **full nonlinear** equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$ (here Λ is the cosmological constant; the no-go invariant $G \cdot \Lambda^2$ below instead uses Λ for the inverse-length UV cutoff — a distinct object) — the point-wise null relation $2 T_{kk} = R_{kk}$ for the actual Ricci tensor, checked nonlinearly — and its **spin-2 (traceless)** part follows as a one-step geometric identity: a symmetric tensor annihilating every null contraction is pure-trace ($S_{ab} k^a k^b = 0 \implies \text{null } k \implies S_{ab} = \frac{1}{2} g_{ab}$, the constraint having rank 9 on the 10 symmetric components). This is **[DERIVED]**, modulo three named gates — the temperature value (axiom R), Jacobson’s local-equilibrium premise $\phi = 0$, and one continuum focusing input. On the negative side we prove a **no-go**: the records cannot fix the absolute scale A (length-weight -2 , in bijection with Newton’s G via $G \cdot A = 1/4$). Under the length-unit gauge $1 \rightarrow 1$, every intrinsic record functional is weight-zero while A is weight -2 ; the contradiction is exact (weight-counting lemma), and the SIGMA-SPLIT construction makes it concrete — the *same* sealed horizon data are consistent with $A_{\text{rec}} = 1$ or $A_{\text{rec}} = 3$, giving the schematic Clausius coefficient $\beta = 1/A = 1$ or 3 with the Einstein residual 0 for both. Both $\kappa \cdot A = 2$ (the physical coupling $\kappa = 8\pi$) and $G \cdot \Lambda^2$ (with Λ the inverse-length UV cutoff, distinct from the unimodular Λ) are fixed weight-zero invariants (separate pure numbers, not numerically equal), each leaving the absolute scale free. The entire no-go rests on one isolated premise — **gate G1**: no sealed law consumes the record area outside the labeling map — exactly as the temperature value rests on axiom R; its sharpest candidate counterexample, an intrinsic seal rate, closes rather than opens it. We show the obstruction

is the same structural fact as QCD's dimensional transmutation (a scale-free record ontology predicts every dimensionless ratio and needs one external unit), and we separate the *derived* spin-2 equation from the *obstructed* propagating graviton, which would require a continuum (type III) structure the finite record lattice cannot host. The result places sealed-record gravity squarely in the emergent/thermodynamic tradition: **gravity's form for free, its scale and its graviton provably not.**

1. What is derived: the equation of state

1.1 Temperature [DERIVED, mod axiom R — established in the horizon-thermodynamics paper of this batch]

The thermodynamic derivation begins with a temperature. The horizon-thermodynamics paper of this batch establishes it two ways on record lattices: the Euclidean cone is smooth at the unique period $\beta = 2/\alpha$ (Gibbons–Hawking regularity, with Bessel-mode matching), and the wedge-reduced record vacuum exhibits Bisognano–Wichmann modular structure $\beta_k = 2/\beta_{k\text{boost}}$ uniformly across resolvable modular levels. We take $T = \alpha/2$ as input. Its value $\beta = 2$ rests on one named axiom (R) — the Euclidean-rotation / modular-period identification, the residue of Bisognano–Wichmann — and we carry that conditional forward without re-deriving it here. As a high-precision consistency check, the accelerated KMS detailed-balance ratio $F(E)/F(-E) = e^{-2E/a}$ holds identically at $T = \alpha/2$ (residual 0 at $\text{dps} = 100$), and the Rindler interval identity $\Delta t^2 - \Delta x^2 = (4/\alpha^2) \sinh^2(\alpha \Delta / 2)$ is exact.

1.2 Geometry-universal area density [DERIVED]

The pure-geometry content of the area law is geometry-universal. A 3+1 massless scalar partial-wave-decomposes into radial chains, and the ball entropy $S(R) = \sum_{l=0}^{2l+1} S_l(R)$ gives an area coefficient that Richardson-extrapolates to **Srednicki's 0.295** [5] with no fitted tail — a genuine universal constant (the spherical-ball partial-wave computation is the program's entanglement-capacity calculation). The ball/planar capacity ratio is $U_G = 1$ to $\sim 10^{-3}$. Crucially, the absolute scale β_A *cancels* in this ratio ($\beta = \beta_A \cdot f_{\text{geom}}$, $U_G = f_{\text{sph}}/f_{\text{pl}}$), so the *geometry factor* f_{geom} is record-derived while the overall scale is not. Two caveats are kept honest: $U_G = 1$ *straddles* unity at the $\sim 10^{-3}$ level (the sign of $U_G - 1$ is cutoff-limited, a consistency not a sign-determination), and flat-screen universality is *measured to break in the angular sector* (the partial-wave first-law ratio departs from unity for $l \geq 1$). So [DERIVED] covers the area-coefficient ratio and the scalar ($l = 0$) channel; the angular sector is a measured departure.

1.3 Pure-area Wald–Noether charge [DERIVED, structural]

The record entropy is screen-cell counting, $S = \beta_A \cdot A + o(A)$, with curvature pieces explicitly subleading. The Wald density is then $s = \beta_A$ with $ds/dR = 0$ **by construction** — the screen-cell capacity does not couple to a curvature scalar — so this is the *statement* that β_A is curvature-free, a structural property of the screen-cell entropy ontology corroborated by the floor-converged mode-cutoff sweep (the apparent curvature-running shrinks monotonically with the cutoff, consistent with a truncation artifact rather than a Wald coupling). This is **not** a

derivation that an arbitrary higher-curvature gravitational action generates no Wald corrections; it is the statement that *this* ontology's entropy is pure-area. The consequence is that $\phi = 1/4G$ is well-defined (not running), keeping the Clausius route in the equilibrium (pure-Einstein) branch rather than the nonequilibrium $f(R)$ /Lovelock branch.

1.4 The field equations in form [DERIVED, mod R + local-equilibrium + focusing gate]

With a pure-area entropy at temperature $T = a/2$, two closures deliver the Einstein equation in form.

Clausius (nonlinear). The Jacobson construction on local Rindler horizons is exact: Raychaudhuri gives $\dot{\theta} = -R_{kk} + O(\ell^2)$, the heat- and area-side affine moments cancel, and the pointwise null relation $2 T_{kk} = R_{kk}$ holds for the **actual Ricci tensor** — checked nonlinearly on explicit Schwarzschild and on an anisotropic fluid with genuine traceless stress (corpus computations), with contracted Bianchi closing. The coefficient match $2/\phi = 8G$ forces $\phi = 1/4G$. This is the **full nonlinear** equation $G_{ab} + \Lambda g_{ab} = 8G T_{ab}$, not a linearized truncation.

Modular first law (linearized) and the spin-2 part. The relative-entropy first law $\delta S = K$ (with $S_{rel} = 0$ strictly second order) yields the linearized equation. Its traceless (spin-2) part follows as a one-step geometric identity: the null-cone lemma — *a symmetric tensor S_{ab} with $S_{ab} k^a k^b = 0$ for every null k is pure-trace, $S_{ab} = \phi g_{ab}$* — fixes **all 9 traceless components at once** (the constraint over a spanning set of null k has rank 9 on the 10 symmetric components; sympy). The same geometric content underlies Jacobson's entanglement-equilibrium derivation; the correct timelike statement of it is that a symmetric E_{ab} whose contraction $E_{ab} u^a u^b$ is the *same* for every unit timelike u is $c g_{ab}$ (the naive vanishing condition $E_{ab} u^a u^b = 0$ unit timelike u instead forces $E = 0$, and is *not* the lemma). The traceless projection removes the free proportionality coefficient c (the cutoff scale Λ), so the **shape** of the spin-2 equation is scale-free; G itself remains explicitly, as the coupling on the right-hand side $8G T_{ab}$.

Conditionality. Three named gates carry the form: axiom (R) (§1.1); Jacobson's **local-equilibrium premise** $\phi = 0$ at the bifurcation — *presumed, not derived* — which selects pure Einstein over the nonequilibrium branch; and the continuum focusing input $\dot{\theta} = -R_{kk}$. None of these bears on the no-go of §2.

1.5 The unimodular fork [DERIVED]

Trace-free Einstein $R_{ab} - \frac{1}{2}g_{ab} R = 8G(T_{ab} - \frac{1}{2}g_{ab} T)$ is a genuine 9-component equation in $d = 4$; its divergence plus contracted Bianchi gives, with $\Lambda := (R + 8G T)/4$, exactly $\nabla_b \Lambda = 8G \nabla_b$ (residual identically 0, sympy) — the conservation/Bianchi obstruction *rerouted* into a dynamical Λ , not evaded, with the *same* $8G$ (no second coupling). The cosmological constant becomes a non-sourced integration constant Λ , a boundary datum.

So sealed-record gravity reaches the Einstein field equations **as the thermodynamics of the records**, deriving more than the vanilla thermodynamic program assumes (the geometry-universality and the pure-area property are derived here, not posited). The remainder establishes precisely where, and why, the path stops.

2. The no-go on Newton's constant [NO-GO]

2.1 The result and its single premise

The records do not, and provably cannot, fix the absolute scale Λ — which has length-weight -2 and is in bijection with Newton's G (weight $+2$) via $G \cdot \Lambda = 1/4$ (so fixing one fixes the other; they are linked, never equal). Under the length-unit gauge $g_- : 1 \rightarrow 1$ holding all sealed data fixed, the records carry only weight-zero (dimensionless / ratio / conformal) invariants. The degree map $\text{weight}(X) = \log[X(1)/X(1)]/\log$ gives, sympy-exact (and numerically at ≈ 1.7 , residual 0):

$\text{weight}(A_{\text{rec}} = 1^2) = +2, \text{weight}(\Lambda = 1/1^2) = -2, \text{weight}(G) = +2, \text{weight}(1_{\text{step}}) = +1, \text{weight}(\text{texture count}) = 0,$

with the bijection $G \cdot \Lambda$ of weight 0 and value $1/4$. Both

$\Lambda^2 = 2$ and (separately) $G \cdot \Lambda^2 = \text{const}$

are fixed weight-zero invariants — *each* a pure number, **not numerically equal to one another** ($\Lambda^2 = 2$ follows from $\Lambda = 8G$ and $G \cdot \Lambda = 1/4$; $G \cdot \Lambda^2$ is a different pure number set by whichever route computes it). What they share is the *structure*: a weight-zero combination is fixed while the absolute scale Λ/G is a free modulus. (Two normalizations of the Clausius coefficient appear: the physical $\Lambda = 8G$ in $\Lambda^2 = 2$, and the schematic $\Lambda_J = 1/\Lambda$ in the SIGMA-SPLIT of §2.2, where $\Lambda_J \cdot \Lambda = 1$. Throughout, Λ is the inverse-length UV/spectral cutoff, distinct from the unimodular cosmological constant Λ of §1.5.)

The single load-bearing premise — gate G1. The no-go rests on exactly one premise, isolated as cleanly as axiom (R) isolates the temperature value: **gate G1** — *no sealed law consumes the record area A_{rec} (a weight- $\pm k$ dimensional datum) except through the continuum labeling map*. G1 is what makes the corpus factor as (R, Λ) with g_- the identity on the record sector R , so that every intrinsic record functional factors through R , is g_- -invariant, and is therefore weight-zero — while Λ is weight -2 . The no-go is airtight *iff* G1 holds. Its sharpest candidate counterexample, an intrinsic *seal rate* Γ (divisions per proper time), **closes** rather than opens it: a record-internal Γ is divisions-per-seal $= 1$ (weight 0, no scale); any geometric Γ is the seal density $1/1_{\text{step}}$ (weight -1 , the gauge/spacing direction itself); and a putative weight- (-2) $\Gamma^2 = (\#^2)/A_{\text{rec}}$ is a free record number times $1/A_{\text{rec}}$ (a relabeling, never the absolute Λ) — an intrinsic absolute scale would need weight -2 *and* weight 0 at once, which the factorization forbids. So the seal-spacing length is provably the unique dimensionful primitive and the sole gauge direction.

2.2 SIGMA-SPLIT: the gauge orbit made concrete [NO-GO]

The mechanism is visible in one construction. The *same* sealed horizon data are consistent with two screen-cell normalizations, $A_{\text{rec}} = 1$ or $A_{\text{rec}} = 3$, giving

$A_{\text{rec}} = 1 \quad \Lambda = 1, \quad \Lambda_J = 1, \quad G = 1/4; \quad A_{\text{rec}} = 3 \quad \Lambda = 1/3, \quad \Lambda_J = 3, \quad G = 3/4,$

with $\Lambda_J = 1/\Lambda$ the schematic Clausius coefficient, $G \cdot \Lambda = 1/4$ invariant in both, and the Einstein-form residual exactly 0 in both (sympy). The absolute G is free; only the gauge-invariant product is fixed. A scale-rescaling check confirms it: $1 \rightarrow 1$ sends $G \rightarrow \Lambda^2 G$ while *every* sealed, order, and modular observable is invariant.

2.3 Every lever collapses the same way [NO-GO]

Each candidate route to fix G reduces to the same weight-zero invariant. A mass-gap correlation length is $(\lambda\#) \cdot l_{\text{step}}$ (lattice units, not an independent length); a second marginal sector transmutes a *dimensionless* fixed-point coupling, tying sectors to one another, never to a geometric length; an asymptotic-safety fixed point $g_* = G^2$ gives $G = g_* l_{\text{step}}^2$, reproducing SIGMA-SPLIT; the conformal trace anomaly (a , c) fixes dimensionless higher-curvature couplings only; Sakharov-induced gravity gives $G \cdot \Lambda^2 = 192^2/N$ with Λ cancelling. The **de Sitter** route is the sharpest test and still collapses: the unimodular Λ is itself weight -2 (the weight-twin of A), and the entropy $S_{\text{dS}} = 1/(G \cdot \Lambda) = N_{\text{dS}}$ (prefactor schematic — the textbook value is $3/(8G\Lambda)$; the weight argument is prefactor-independent) fixes only the weight-zero product $G \cdot \Lambda$, with $N_{\text{dS}} = A_{\text{horizon}}/A_{\text{rec}} = 4(A/\Lambda)$ the ratio of two weight- (-2) data — 1-independent (sympy $d/dl = 0$), so it adds no gauge-eligible datum and G stays free. Even the strongest *record-unit* attack fails: measuring a mass in record units smuggles in the record-length lab-length conversion that g_* acts on. The unification is the content: *deriving $1/G$ always trades it for a labeling-equivalent cutoff; the missing datum is exactly one absolute length unit, which no weight-zero record functional can be.*

2.4 The same fact as dimensional transmutation

The no-go is not a pathology; it is the gravity-sector face of a familiar fact. A confining gauge theory predicts every dimensionless mass *ratio* and the running of a dimensionless coupling, but **dimensional transmutation does not produce an absolute scale** — Λ_{QCD} in physical units is an external input, and a lattice mass gap comes out as a pure number times the lattice spacing. Sealed-record gravity says the same of G : the records fix every dimensionless ratio and the Einstein *form*, and need one external unit to set the scale. What the no-go forbids is *one absolute dimensionful unit* — **not** the dimensionless gravitational coupling-per-species $c_m = Gm^2/c$, which is weight-zero, intrinsic, and therefore *eligible* to be a record output (the open hierarchy question, the gravity-sector analog of predicting dimensionless mass ratios). A theory that reconstructed gauge physics up to one scale yet fixed G absolutely would be the contradiction; the no-go is the program being consistent with itself.

3. The tensor sector: the spin-2 equation is derived, the graviton is not

The linearized spin-2 Einstein *equation* $G_{ab} = 8G T_{ab}$ is derived geometrically (§1.4). What is **[OBSTRUCTED]** is the *distinct* claim that the matter modular *currency* independently **prices** the traceless stress T_{ij} as a universal first-law charge — i.e. the propagating graviton / quantum spin-2 operator, *not* the equation. The modular currency prices the 4 energy-momentum components (T_{00} , T_{0i}) at a universal capacity $1/(4)$; a congruence of boosted wedges does **span** the 5 traceless spatial-stress components (rank 5), but only at a **non-universal, boost-dependent, second-order** coupling $\cdot \sinh^2$, never the single universal capacity that prices energy and momentum.

Four obstructions converge on **one root**. (i) *Non-universality*: the traceless coupling is a second-order susceptibility, not a capacity. (ii) *Type I, no area operator*: the finite record

lattice is a type-I algebra with no center-valued area operator; the crossed-product constructions that manufacture a fluctuating area operator from an algebra lacking one are blocked — the modular flow is inner/trivial, and the entropy-production (arrow) flow, though genuinely outer, is a *stationary current* preserving an invariant measure, so its crossed product is semifinite (type I, c-number area), not the type III needed. (iii) *Berry*: the first-order relative modular charge between boosted wedges has zero spin-2 ($\cos 2$) overlap. (iv) *Null-cut*: the universal-2 null-cut pricing is a half-sided modular inclusion $[K, P] = iP$, which has no finite-dimensional representation. The single root: **every spin-2 / quantum-area route needs an outer modular flow, a type III algebra, and a half-sided inclusion — structures that live only in the continuum local algebra, which the finite record lattice cannot host intrinsically.** So sealed-record geometry is spin-2-*active* but not-a-graviton; a genuine propagating spin-2 (the real-time transverse-traceless sector, an emergent holographic bulk) is gated and remains [OPEN].

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4. The open frontiers, and the emergent/intrinsic distinction

Two frontiers remain open; neither bears on the no-go of §2.

Covariance of the dynamics. [OBSTRUCTED-leaning] Done the causal-set way, discreteness covariantizes the *kinematics* outright: a Poisson sprinkling of division events is statistically Lorentz-invariant; the entropy-production arrow $= D(P_{AB}||P_{BA})$ reads the frame-invariant causal partial order, not a preferred slice; and the free-flash localization dissolves Hegerfeldt's obstruction (a point-event sampled from a density, not a positive-energy state). The interacting-*distinguishable* dynamics is solved (Tumulka 2020 [12]). The genuine residue is a covariant interacting-*field* (variable particle number) record dynamics, which remains [OBSTRUCTED-leaning].

The propagating graviton and the real-time dynamics. [OPEN] Beyond the static modular-currency program lies the real-time transverse-traceless sector — the radiative, propagating content of gravity, whose quantum is the graviton. The split is sharp: an *intrinsic finite-record* graviton is obstructed (§3, the type-I/type-III wall), while an *emergent continuum* graviton — gravity as an effective field theory of the coarse-grained metric, reached in a scaling limit — is open, not obstructed, and is the same continuum-limit program as reconstructing continuum field theory from the finite lattice. So the records carry the *equation of state* of gravity, and the *dynamics* of gravity (its radiative, propagating sector) is the emergent-continuum-limit frontier, not an intrinsic finite-record object.

The observability counterpart of these frontiers — whether the gravitational record process is genuinely indivisible, and whether that is measurable — is treated in the companion decoherence paper of this batch, which proves it is structural and not certifiable from the reduced channel.

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5. Status ledger

Object	Status	Statement
Temperature $T = s/2$	[DERIVED, mod R]	established in the horizon paper; value rests on axiom (R)
Geometry factor f_{geom} (area coeff. + 1=0)	[DERIVED]	$U_G = 1$ to -10^{-3} (straddles); Srednicki 0.295, no fit
Angular first-law universality	[OPEN]	departs for $1 - 1$; measured, not closed
Horizon entropy = pure-area Wald charge	[DERIVED, structural]	$ds/d\Lambda = 0$ by construction; corroborated by the floor-converged sweep
Clausius / modular first law $\rightarrow$ Einstein form	[DERIVED, mod R + $=0$ + focusing]	full nonlinear via Clausius; linearized via $S = K$; $\rightarrow 1/4s$
Cosmological constant Λ (unimodular)	[DERIVED]	non-sourced integration constant; Bianchi rerouted
Absolute scale $\Lambda \cdot G$	[NO-GO]	$\cdot \Lambda$ and $G \cdot \Lambda^2$ fixed weight-zero invariants, scale free; weight-counting lemma; SIGMA-SPLIT; single premise gate G1 (seal-rate attack closes it); at audited scope
Linearized spin-2 (traceless) Einstein equation	[DERIVED, mod R + gates]	null-cone lemma fixes all 9 traceless comps; traceless shape scale-free (G remains on the RHS)
Spin-2 matter charge / propagating graviton	[OBSTRUCTED $\rightarrow$ OPEN]	non-universal + type-I + Berry + half-sided-inclusion; real-time TT sector gated
Covariant interacting field dynamics	[OBSTRUCTED-leaning]	kinematics covariantized; interacting-distinguishable solved; field case open

6. What this paper does and does not claim

It **claims**: a derivation, from the sealed-record ontology, of the gravitational field equations as an equation of state (temperature + pure-area geometry-universal entropy + modular/Clausius first law), including the full nonlinear form and the linearized spin-2 (traceless) equation as a geometric identity, modulo the three named gates; and a no-go (at audited corpus scope, with gate G1 as its single load-bearing premise) that the absolute coupling G is not encoded in the records ($\cdot \Lambda$ and $G \cdot \Lambda^2$ weight-zero invariants; the weight-counting lemma; SIGMA-SPLIT).

It does **not** claim: a derivation of Newton's G (proved impossible, modulo the audit scope); a propagating graviton / quantum spin-2 charge (obstructed — the type-I/type-III wall — and distinct from the derived equation); the real-time/radiative dynamics of gravity (open, the emergent-continuum frontier); a covariant interacting-field dynamics (open, with the kinematics covariantized); or the matter-sector hierarchy c_m (a separate, intrinsic, still-open dimensionless quantity the no-go does *not* forbid). The framing is thermodynamic: gravity as the equation of state of records, with the entropy-area density a calibration — and the theorem (at audited scope) that the missing datum is exactly *one absolute unit*. The honest one-line summary: **the records give gravity's form for free and prove they cannot give gravity's scale.**

7. Precision discipline

Every modular-kernel, near-vacuum, and weight-bookkeeping quantity was computed in mpmath at `dps 80` (commonly 100), with exact algebra (the weight degree map, SIGMA-SPLIT, the null-cone lemma, the trace-free / Bianchi identities, the de Sitter 1-cancellation, the KMS detailed-balance and Rindler identities) in sympy. The receipt script `code/v6_pXI_sealed_record_gravity_nogo_receipts.py` reproduces the no-go and equation-of-state numbers below (the §1.2 geometry-universality constants — Srednicki's 0.295 [5] and the U_G ratio — are the corpus's spherical-ball partial-wave computation, separate from this receipt): the weight map (residuals 0), the bijection $G \cdot \Lambda = 1/4$ (weight 0), SIGMA-SPLIT ($= 1$ vs 3, residual 0), the two separate invariants $\cdot \Lambda = 2$ and $G \cdot \Lambda^2$, the de Sitter weight-twin (N_{dS} 1-independent), the null-cone lemma (rank 9; pure-trace solution) together with the correct-versus-naive timelike statement, the spectral-action honesty check (the naive collar profile gives $f = 1/(2)^2 = 0.025330\dots$, *not* $1/(2)$, so $G \cdot \Lambda^2 = 3^2$ does not follow), and the KMS/Rindler identities. Never float64 for the modular-kernel or near-vacuum quantities, where the corpus audit shows a 25–32% artifact.

References

Companion papers.

[V] *Quantum theory from sealed records I–II*, companion papers ([paper-Va-foundations-1](#), [paper-Vb-foundations-2](#)) — axioms R, S, C; the $q = 2$ weight calculus; Lorentz signature from commitment order. [VIII] *Horizon thermodynamics on record lattices*, companion paper ([paper-VIII-horizons](#)) — temperature two ways (Euclidean cone $= 2 /$ and Bisognano–Wichmann modular structure), censorship, greybody impedance, capacity; the input for §1.1. [X] *Gravitational decoherence does not certify its mechanism*, companion paper ([paper-X-gravitational-decoherence](#)) — the operational-undecidability counterpart of the open covariance/indivisibility frontier of §4.

External.

[1] T. Jacobson, *Thermodynamics of spacetime: the Einstein equation of state*, Phys. Rev. Lett. **75**, 1260 (1995). [2] T. Jacobson, *Entanglement equilibrium and the Einstein equation*, Phys. Rev. Lett. **116**, 201101 (2016). [3] C. Eling, R. Guedens, T. Jacobson, *Nonequilibrium thermodynamics of spacetime*, Phys. Rev. Lett. **96**, 121301 (2006). [4] R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48**, R3427 (1993). [5] M. Srednicki, *Entropy and area*, Phys. Rev. Lett. **71**, 666 (1993). [6] A. D. Sakharov, *Vacuum quantum fluctuations in curved space and the theory of gravitation*, Sov. Phys. Dokl. **12**, 1040 (1968). [7] T. Padmanabhan, *Thermodynamical aspects of gravity: new insights*, Rep. Prog. Phys. **73**, 046901 (2010). [8] E. Verlinde, *On the origin of gravity and the laws of Newton*, JHEP **04**, 029 (2011). [9] H.-J. Borchers, *On revolutionizing quantum field theory with Tomita’s modular theory*, J. Math. Phys. **41**, 3604 (2000). [10] V. Chandrasekaran, R. Longo, G. Penington, E. Witten, *An algebra of observables for de Sitter space*, JHEP **02**, 082 (2023). [11] N. Lashkari, J. Lin, H. Marrochio, R. Myers, R. C. Myers — *modular/relative-entropy first law and gravitational dynamics* (entanglement-equilibrium and JLMS lineage). [12] R. Tumulka, *A relativistic version of the GRW flash model*, J. Stat. Phys. **125**, 825 (2006); and the interacting-distinguishable extension (2020).